

Introduction to `curl` for API Testing

A quick guide to using `curl` for testing REST APIs from the command line.

✓ What is `curl`?

- Command-line tool to **send HTTP requests**
- Supports many protocols, mainly HTTP/S
- Excellent for **API testing and debugging**

Basic Syntax

```
curl [options] [URL]
```

Examples are usually focused on RESTful HTTP APIs.

1. GET Request

Retrieve data from an API.

```
curl https://jsonplaceholder.typicode.com/posts/1
```

- Fetches post with ID = 1

2. POST Request

Send JSON to create data.

```
curl -X POST https://jsonplaceholder.typicode.com/posts \  
  -H "Content-Type: application/json" \  
  -d '{"title": "foo", "body": "bar", "userId": 1}'
```


3. PUT Request

Update existing resource.

```
curl -X PUT https://jsonplaceholder.typicode.com/posts/1 \  
  -H "Content-Type: application/json" \  
  -d '{"id": 1, "title": "updated", "body": "new body", "userId": 1}'
```


4. DELETE Request

Delete a resource.

```
curl -X DELETE https://jsonplaceholder.typicode.com/posts/1
```


5. Include Response Headers

Use `-i` to show response headers.

```
curl -i https://jsonplaceholder.typicode.com/posts/1
```


6. Send Bearer Token

Use `Authorization` header for secured APIs.

```
curl -H "Authorization: Bearer YOUR_TOKEN_HERE" \  
https://api.example.com/protected
```


Summary

- `curl` is a versatile HTTP client for API testing
- Supports GET, POST, PUT, DELETE
- Use headers, data, and auth for realistic testing